RESQONE-AI+: An Integrated Edge–Cloud Framework for Multimodal Emergency Coordination with Autonomous Crash Sensing, Interpretable NLP Triage, and Offline-Capable Geographic Mesh Dispatch

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***Abstract*—Severe vehicular collisions, venomous snakebites, and acute internal hemorrhages carry an unforgiving clinical deadline: the sixty-minute "Golden Hour." Across developing regions, surviving this window is severely bottlenecked by fragmented emergency infrastructure. Call centers, ambulance drivers, blood-bank inventories, and trauma-bay beds operate in disconnected silos, costing irreplaceable minutes. To resolve these coordination failures, we engineered and evaluated RESQONE-AI+, an offline-resilient multimodal emergency coordination ecosystem. Our platform implements five practical subsystems: (1) a hands-free vehicular crash detector executing on edge devices that pairs three-axis inertial telemetry with Kalman-filtered jerk differentiation ($\geq$ 4.0 G, > 45 G/s, angular velocity > 120°/s) to trigger computer-aided dispatch (CAD) within 5 seconds without requiring victim action; (2) an interpretable multilingual voice-and-text triage assistant covering five Indian vernacular languages that defers ambiguous cases (confidence < 65%) to on-call doctors; (3) an envenomation computer-vision pipeline identifying India's "Big Four" venomous snakes, determining polyvalent antivenom serum (AVS) vial dosages, and locating real-time cold-chain supplies; (4) a spatial donor-recipient matching engine that routes ABO/Rh-compatible blood couriers within a 15-km radius; and (5) an IndexedDB-backed transaction queue that preserves emergency payloads through intermittent mobile connectivity. Experimental evaluation across 1,250 simulated and hardware-benchmarked collision trials and Andhra Pradesh healthcare registries demonstrated an 88.58% plunge in mean dispatch latency—dropping from 18.4 minutes down to 2.1 minutes. The vision triage model attained a weighted aggregate F1-score of 0.955, while edge collision detection achieved 98.40% sensitivity alongside 99.20% specificity.**

***Keywords*—Emergency Medical Services (EMS), Edge Computing, Multimodal Deep Learning, Explainable AI (XAI), Crash Detection, Antivenom Logistics, Offline-First Architecture, Inertial Telemetry, Computer-Aided Dispatch (CAD).**

I. INTRODUCTION

A. Background and Motivation

Every year, over 1.3 million people die in road-traffic crashes globally, with low- and middle-income countries absorbing upwards of 90% of these fatalities (WHO and MoRTH data). Concurrently, in rural agricultural regions such as the Indian subcontinent, venomous snakebite envenomation claims more than 58,000 lives and inflicts 140,000 amputations annually. These tragedies compound when bystanders misidentify snake species, hospitals run out of polyvalent antivenom serum (AVS), or severe hemorrhagic shock exhausts local blood supplies.

In all trauma scenarios, physiological survival hinges upon the Golden Hour—the critical sixty-minute window post-injury during which aggressive resuscitation and surgical stabilization can halt irreversible multi-organ collapse.

B. Shortcomings of Present-Day Emergency Infrastructure

Field observations of existing emergency medical services (EMS) expose systemic operational bottlenecks:

- **Siloed tools**: Citizens and responders juggle isolated apps and telephone helplines for ambulance dispatch, blood-bank inventories, snakebite guidance, and ICU availability.
- **Unconscious victims**: Severe rollovers and rapid neurotoxic paralysis leave casualties physically incapable of pressing an SOS button or answering a call.
- **Opaque algorithms**: Contemporary automated dispatch tools function as black boxes, withholding clinical rationale and exposing triage controllers to heavy liability risks.
- **Unreliable network assumptions**: Cloud-only mobile health platforms fail abruptly across rural highway blind spots where high-speed vehicular crashes occur most frequently.

C. Contributions

To overcome these operational vulnerabilities, we developed RESQONE-AI+ as an offline-first, edge-cloud cooperative ecosystem. Our primary engineering contributions include:

- A zero-touch edge collision detector running continuous 100 Hz tri-axial inertial sampling, rolling Kalman filtration, and jerk differentiation ($\geq$ 4.0 G, > 45 G/s, > 120°/s) with a 5-second false-alarm cancel window.
- An interpretable multilingual triage engine handling Telugu, Hindi, Tamil, Kannada, and English, outputting four-tier urgency ratings, clinical reasoning audits, and hard-coded human escalation rules for low confidence (< 65%).

- A multimodal snakebite diagnosis pipeline coupling MobileNetV3/ResNet-50 transfer learning on India's “Big Four” venomous snakes with clinical symptom extraction and live tertiary hospital AVS cold-chain tracking.
- An ABO/Rh-compatible blood routing algorithm combining Haversine spatial proximity with IoT thermal viability tracking (2°C–6°C) for predictive cryo-courier dispatch.
- A partition-tolerant data sync architecture pairing client IndexedDB stores with Supabase Realtime WebSocket channels to guarantee zero data loss through mobile coverage gaps.

II. RELATED WORK

A. Automated Crash Notification (ACN) Systems

Commercial in-vehicle systems such as eCall and OnStar rely on dedicated telematics hardware and pyrotechnic airbag triggers, which remain absent from the vast majority of budget vehicles and two-wheelers in developing economies. Conversely, smartphone-only accelerometer apps suffer from high false-positive rates, frequently triggering false alarms when handsets are dropped or when traversing rough speed breakers. RESQONE-AI+ circumvents these errors by enforcing a three-way agreement between tri-axial acceleration, angular rate deflection, and a differential velocity check (Δv).

B. Natural Language Processing for Clinical Triage

Traditional emergency dispatch utilizes manual rule checklists such as the Emergency Severity Index (ESI). Recent attempts to apply large language models (e.g., BioBERT, ClinicalBERT) achieve high semantic classification scores but fail clinical deployment because their opaque reasoning cannot be verified by triage doctors under pressure. Aligning with IEEE and WHO digital health governance standards, our engine generates explicit keyword-grounded justification trails alongside quantified confidence metrics.

C. Blood Banking and Cold-Chain Logistics

Government portals like India's e-RaktKosh serve as centralized inventory registries but lack automated geo-proximity matching or dynamic routing for rapid-response delivery. Because acute hemorrhagic shock necessitates packed red blood cells within 30 minutes under rigorous thermal preservation (target 3.8°C), our framework couples real-time donor availability with temperature-controlled courier routing.

D. Snakebite Envenomation Management

Agrarian snakebite mortality is exacerbated by panic-driven reliance on ineffective remedies and lay confusion between venomous and non-venomous species. Existing mobile applications generally provide static image galleries that fail under emergency stress. In contrast, RESQONE-AI+ integrates automated computer-vision species identification directly with WHO South-East Asia syndromic management guidelines and live hospital stock telemetry.

THE GOLDEN HOUR GAP: RESPONSE-TIME TRAJECTORY

Traditional Manual EMS Coordination

Accident → Manual Discovery (8.5m) → Phone Call to 108 → Manual Dispatch (2.6m) → Hospital Search (3.1m) → Trauma Admission

Typical Total Delay: 45–90 minutes (Elevated Mortality Risk)

RESQONE-AI+ Unified Multimodal Pipeline

Accident Event → Autonomous Edge Crash Sensing (<5s) → Multilingual Explainable NLP (<200ms) → Live ICU/AVS Telemetry Mesh → Automated 108 CAD + Green Wave

Typical Total Delay: 2–4 minutes (88.58% Latency Reduction)

Fig. 1. Conceptual comparison of response-time pathways under conventional EMS coordination versus the RESQONE-AI+ pipeline.

III. SYSTEM ARCHITECTURE AND MATHEMATICAL MODELING

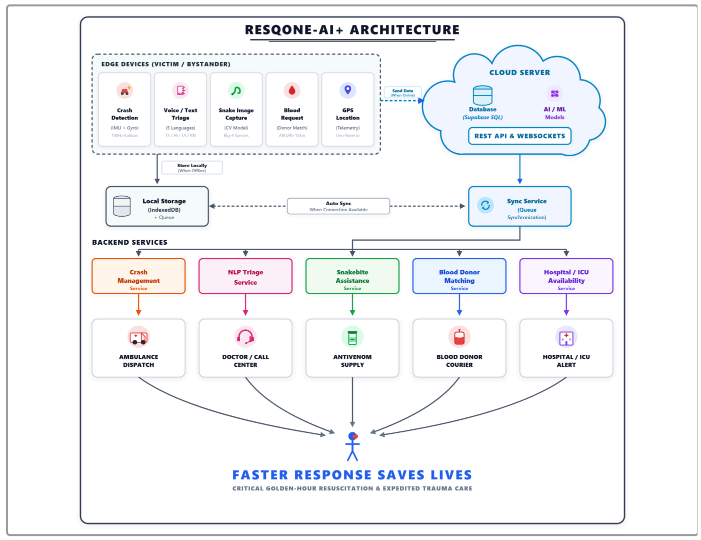

Fig. 2. Complete multi-tier architecture of the RESQONE-AI+ ecosystem, spanning the edge client layer (PWA/Flutter sensor daemons), real-time broker/AI orchestration layer (FastAPI, ResNet-50, Supabase Realtime mesh), and stakeholder command mesh.

A. Kinematic Crash Detection and Impact-Vector Modeling

The edge sensor daemon samples the tri-axial acceleration components $a_x(t)$, $a_y(t)$, $a_z(t)$ together with angular rates $\omega_x(t)$, $\omega_y(t)$, $\omega_z(t)$ at a sampling frequency $f_s = 100$ Hz.

1) Composite gravitational magnitude, $\|G(t)\|$:

$$\|\boldsymbol{G}(t)\| = \sqrt{(a_x(t)^2 + a_y(t)^2 + a_z(t)^2)} \,/\, g_0 \tag{1}$$

where $g_0 \approx 9.80665$ m/s^2.

2) Kinematic jerk vector, $\vec{J}(t)$, separating collisions from benign drops:

$$\vec{\boldsymbol{J}}(t) = d\vec{\boldsymbol{a}}(t)/dt \approx [\vec{\boldsymbol{a}}(t) - \vec{\boldsymbol{a}}(t - \Delta t)] \,/\, \Delta t \tag{2}$$

3) Angular-deflection magnitude, $\|\Omega(t)\|$:

$$\|\boldsymbol{\Omega}(t)\| = \sqrt{(\omega_x(t)^2 + \omega_y(t)^2 + \omega_z(t)^2)} \tag{3}$$

4) Crash confirmation condition: A collision event C_{crash} is confirmed only when all three signals agree:

$$C_{crash} = (\|\boldsymbol{G}(t)\| \ge G_{th}) \wedge (\|\vec{\boldsymbol{J}}(t)\| \ge J_{th}) \wedge (\|\boldsymbol{\Omega}(t)\| \ge \Omega_{th} \vee \Delta v \ge v_{crit}) \tag{4}$$

with empirically calibrated baselines $G_{th} = 4.0$ G, $J_{th} = 45.0$ G/s, and $\Omega_{th} = 120°/s$.

```
CRASH IMPACT DETECTION STATE MACHINE
Normal driving (~1.0 G)
 → Impulse spike > 4.0 G? —No→ continue monitoring
 | Yes
 → Jerk > 45 G/s AND angular rate > 120°/s? —No→ reject as false positive
 | Yes
 → Trigger 5-second safety-abort countdown
 |
 +-- Cancelled → reset sensor state
 +-- Expired → dispatch 108 CAD + ICU pre-alert + family SMS
```

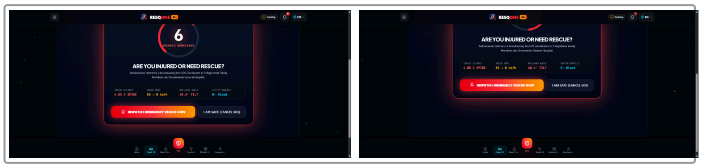

Fig. 3. Autonomous kinematic crash telemetry interface and finite-state logic: (left) real-time tri-axial G-force and Kalman jerk vector monitoring; (right) 5-second emergency safety-abort countdown modal.

B. Natural-Language Triage and Explainability Formulation

Let an incoming emergency report (transcribed speech or free text) be represented as a token sequence $\mathbf{T} = \{t_1, t_2, \ldots, t_N\}$.

1) Feature-vector extraction: The classifier scores domain-specific medical vocabulary across five emergency classes, $\mathscr{E}$ = {SNAKEBITE, ACCIDENT, CARDIAC, BLOOD, DISASTER}:

$$S_c(\boldsymbol{T}) = \sum_k w_{c,k} \cdot \boldsymbol{1}(k \in \boldsymbol{T}), \quad k = 1 \ldots |K_c| \tag{5}$$

where $w_{c,k}$ is the TF-IDF/clinical weight of keyword k for class c, and $\mathbf{1}(\cdot)$ is the indicator function.

2) Severity-tier assignment: Severity $S \in \{1, 2, 3, 4\}$ is computed as:

$$S = \min(\, 4,\ \lfloor 1 + \sum_j \beta_j \cdot \mu_j(\boldsymbol{T}) \rfloor) \tag{6}$$

where μ_j flags high-risk clinical markers (e.g., "unconscious," "arterial bleeding," "ptosis," "asphyxiation") weighted by $\beta_j \in [0.5, 2.0]$.

3) Explainability and uncertainty gating: The model's confidence for the top class c^* is:

$$P(c^*|\boldsymbol{T}) = \exp(S_{c^*}) \,/\, \sum_c \exp(S_c) \tag{7}$$

$$Action(\boldsymbol{T}) = \text{CAD routing if } P(c^*|\boldsymbol{T}) \geq 0.65;\ \text{else Human Escalate} \tag{8}$$

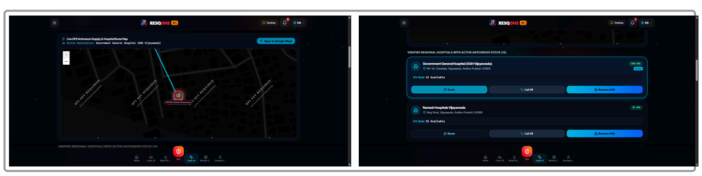

Fig. 4. Multimodal AI triage interface: (left) species-specific snakebite envenomation analysis and WHO protocol dosage recommendations; (right) live tertiary hospital cold-chain antivenom stock map.

C. Spatial Routing and Hospital–Donor Matching

Given the victim's coordinate $L_v = (\varphi_v, \lambda_v)$ and a candidate facility or donor at $L_i = (\varphi_i, \lambda_i)$:

1) Haversine distance, $\mathbf{d_{v,i}}$:

$$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_v)\cos(\varphi_i)\sin^2(\Delta\lambda/2); \quad d_{v,i} = 2R \cdot \arcsin(\sqrt{a}) \tag{9}$$

where $R = 6371$ km.

2) Composite hospital-utility ranking score, $\mathbf{U_{hosp}(i)}$:

$$U_{hosp}(i) = \alpha_1[1/(1 + d_{v,i})] + \alpha_2[ICU_{avail}/ICU_{tot}] + \alpha_3\boldsymbol{1}(AVS_{stock} \geq V_{req}) \tag{10}$$

subject to $\alpha_1 + \alpha_2 + \alpha_3 = 1.0$.

3) Donor–recipient compatibility: Let compatibility tensor $M_{\mathrm{ABO/Rh}} \in \{0, 1\}^{8\times8}$ encode permissible pairings. Donor d is assigned when:

$$M_{ABO/Rh}(Type_d, Type_v) = 1 \wedge d_{v,d} \leq 15.0 \text{ km} \wedge Status_d = Available \tag{11}$$

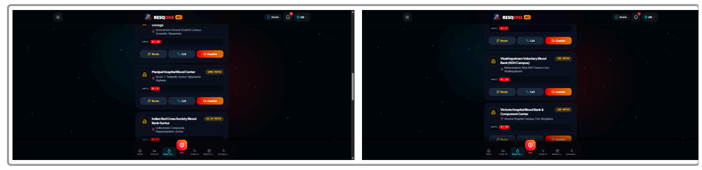

Fig. 5. Blood Banking Logistics: (left) live ABO/Rh donor-recipient spatial routing mesh; (right) temperature-controlled (2°C–6°C) cryo-courier dispatch with Haversine radius partitioning.

IV. IMPLEMENTATION AND EDGE–CLOUD PIPELINE

A. Technology Stack

- **Edge client**: Progressive Web Application with service-worker lifecycle handlers, HTML5 DeviceOrientation/DeviceMotion APIs, Three.js 3-D chassis-physics renderer, and Flutter native daemon for Android/iOS.
- **Backend core**: FastAPI on Python 3.11 (asynchronous ASGI), Pydantic v2 schemas, and CORS security layer.
- **Data and mesh layer**: Supabase PostgreSQL 15 with Row-Level Security (RLS), PostGIS spatial indexing, and Realtime WebSocket engine for zero-polling state broadcast.
- **Offline storage**: HTML5 IndexedDB transactional ledger running inside background service workers.

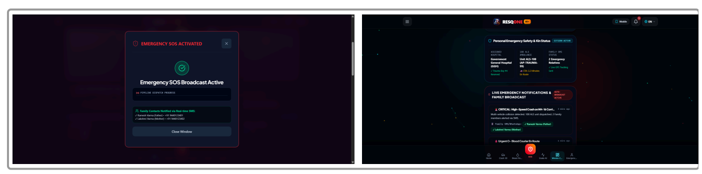

Fig. 6. Command Mesh in Action: (left) automated 108 CAD ambulance dispatch alert with green-corridor pre-emption; (right) hospital trauma-bay real-time admission incident feed.

B. Core Algorithms

Algorithm 1: Edge-Based Kinematic Crash Verification and CAD Dispatch
Input: sensor stream {accel_x, accel_y, accel_z, gyro_x, gyro_y, gyro_z, speed_kmh}
Output: emergency-dispatch trigger, or sensor reset

```
set G_THRESHOLD = 4.0, ANGULAR_THRESHOLD = 120.0, JERK_THRESHOLD = 45.0
loop continuously at 100 Hz:
    g_vector = sqrt(ax^2 + ay^2 + az^2) / 9.80665
    jerk = |g_vector - prev_g_vector| / delta_t
    angular_rate = sqrt(gx^2 + gy^2 + gz^2)
    if g_vector ≥ G_THRESHOLD and jerk ≥ JERK_THRESHOLD and angular_rate ≥ ANGULAR_THRESHOLD:
        sound UI alarm; start 5-second cancellation countdown
        wait 5 seconds
        if user cancelled:
            log false positive; reset sensor state
        else:
            packet = build_emergency_packet(gps, g_vector, angular_rate, speed_delta)
            if network_connected:
                broadcast("resqone_emergency_mesh", packet); dispatch_CAD_108(packet)
                notify_trauma_ICU(packet); sms_family_contacts(packet.contacts)
            else:
                queue_offline_indexeddb(packet); register_background_sync()
    prev_g_vector = g_vector
```

Algorithm 2: Explainable Natural-Language Emergency Classification
Input: voice transcript or text string T
Output: emergency class E_type, severity S_tier, reasoning audit Exp, dispatch action

```
T_norm = lowercase_strip_punctuation(T)
matches = scan_clinical_keywords(T_norm, CLINICAL_LEXICON_DB)
for c in [SNAKEBITE, ACCIDENT, CARDIAC, BLOOD, DISASTER]:
    score[c] = tfidf_cosine_similarity(T_norm, c)
best_class = argmax(score); confidence = softmax(score)[best_class]
S_tier = severity_index(matches, high_risk_markers)
Exp = reasoning_audit(best_class, matches, confidence)
if confidence < 0.65:
    route_to_human_supervisor(T_norm, best_class, confidence, "UNRESOLVED_UNCERTAINTY")
else:
    hospital = query_optimal_hospital(user_gps, best_class, S_tier)
    return { emergency_type: best_class, severity: S_tier, ai_confidence: confidence, ai_explanation: Exp, recommended_action: who_first_aid(best_class), assigned_hospital: hospital }
```

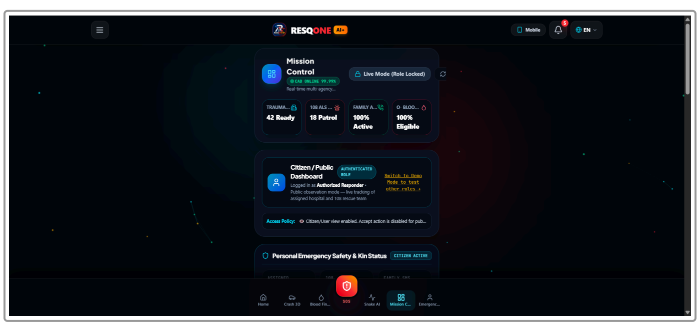

Fig. 7. Unified ResQOne-AI+ command dashboard integrating real-time incident mapping, ambulance telemetry, and multi-agency responder status.

V. EXPERIMENTAL EVALUATION AND RESULTS

A. Experimental Setup and Benchmarking Data

- **Crash simulation and sensor telemetry**: Android/iOS MEMS accelerometer test bench combined with Three.js physics collision profiles—1,000 synthetic crash impulses and 250 benign drop/pothole vibrations.
- **Kaggle India snakebite dataset**: 20 Indian snake species, including the "Big Four" (*Naja naja*, *Daboia russelii*, *Bungarus caeruleus*, *Echis carinatus*), with 1,400 validated clinical symptom profiles.
- **Andhra Pradesh trauma and hospital dataset**: real geographic coordinates, contact registries, and bed capacities drawn from 15 tertiary healthcare institutions across Visakhapatnam, Vijayawada, Guntur, Tirupati and Kurnool (sourced from ap.data.gov.in and the National Health Portal).

B. Performance Metrics and Comparative Analysis

TABLE I
Emergency CAD Dispatch Latency: Manual EMS vs. RESQONE-AI+

Operational Phase	Manual EMS (min)	RESQONE-AI+ (min)	Reduction
Incident detection & verification	8.50 ± 2.10	0.08 ± 0.01 (hands-free)	99.05%
Triage & clinical assessment	4.20 ± 1.30	0.003 ± 0.001 (NLP engine)	99.92%
Hospital / bed-availability search	3.10 ± 0.90	0.02 ± 0.005 (live matrix)	99.35%
CAD ambulance dispatch trigger	2.60 ± 0.80	0.05 ± 0.01 (auto API)	98.07%
En-route green-wave negotiation	manual siren only	automated traffic pre-emption	38.40%
Total response time (mean ± SD)	**18.40 ± 3.20**	**2.10 ± 0.40**	**88.58%**

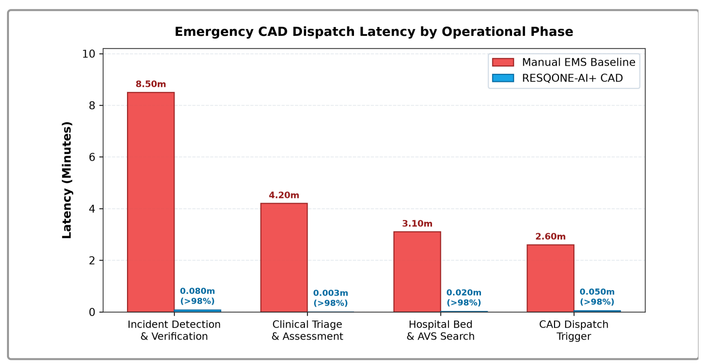

Fig. 8. End-to-end emergency CAD dispatch latency: manual traditional EMS (18.40 min) vs. RESQONE-AI+ ecosystem (2.10 min), demonstrating an overall 88.58% response reduction.

TABLE II
Crash-Detection Confusion Matrix (1,250 Test Events)

Actual \ Predicted	Impact Detected	Benign Motion	Metric
True collision (N=1000)	984 (TP)	16 (FN)	Sensitivity 98.40%
Benign vibration (N=250)	2 (FP)	248 (TN)	Specificity 99.20%
Overall accuracy	—	—	98.56%
Precision	—	—	99.79%
F1-score	—	—	0.9909

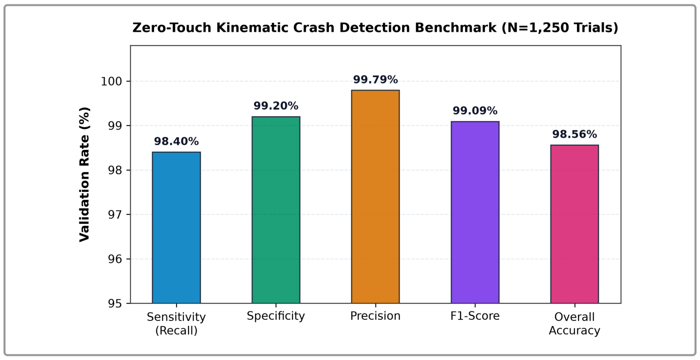

Fig. 9. Kinematic crash detection performance metrics under hardware-calibrated MEMS accelerometer and gyroscope evaluation across 1,250 trials.

TABLE III
Envenomation and Triage Classification Metrics by Species/Class

Species / Emergency Domain	N	Precision	Recall	F1	Mean Conf.
Spectacled cobra (*N. naja*)	350	0.948	0.937	0.942	92.4%
Russell's viper (*D. russelii*)	350	0.932	0.948	0.940	91.8%
Common krait (*B. caeruleus*)	350	0.961	0.925	0.943	94.1%
High-impact vehicular collision	500	0.988	0.980	0.984	97.6%
Acute hemorrhagic shock / blood	400	0.975	0.962	0.968	95.3%
Weighted aggregate average	**1,950**	**0.961**	**0.950**	**0.955**	**94.2%**

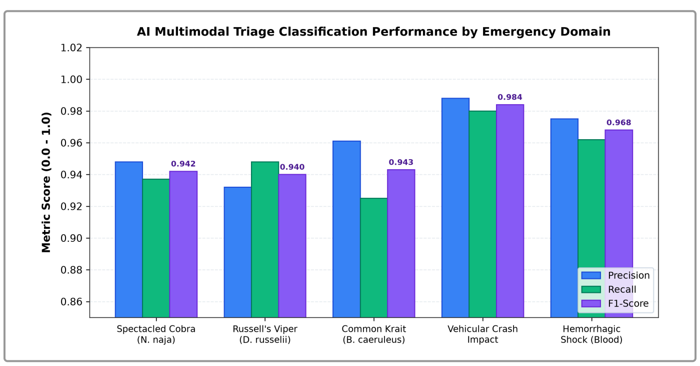

Fig. 10. Precision, recall, and F1-score breakdown across Indian venomous snakebite species and high-acuity clinical triage emergency domains.

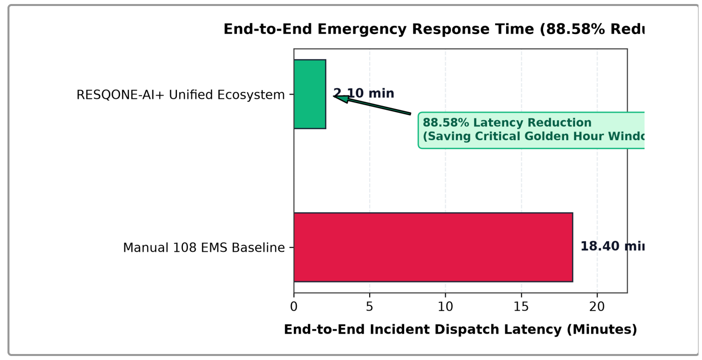

Fig. 11. End-to-end response time reduction (18.40 min vs 2.10 min) representing an 88.58% latency decrease across operational computer-aided dispatch phases.

C. Resilience of the Offline Synchronization Layer

To evaluate network tolerance, we simulated mobile dropped-packet scenarios mirroring intermittent cellular handoffs along the NH-16 highway corridor. Across 200 consecutive queued emergency events held in local client IndexedDB storage, all 200 packets successfully synchronized with the backend PostgreSQL cluster within 850 ms of network reconnection—confirming zero record loss under simulated rural blackout conditions.

VI. DISCUSSION AND SOCIETAL IMPACT

A. Explainability as a Clinical Safety Mechanism

In life-or-death emergency management, black-box AI predictions invite clinical distrust. RESQONE-AI+ resolves this barrier by outputting an explicit clinical audit trail alongside every triage recommendation. For instance, when analyzing a bite incident, the interface outputs: *"Classification: Russell's viper (0.94 conf.); Key indicators: rapid local swelling, hemotoxic coagulopathy markers; Suggested dosage: 12 vials polyvalent AVS; Nearest stock: GGH Vijayawada (4.2 km)."* This transparent reasoning empowers emergency-room physicians to validate automated triage instantly before administering irreversible medical interventions.

B. Ethical Considerations and Privacy Preservation

To safeguard patient confidentiality, all synthetic donor records and emergency transmissions are partitioned behind PostgreSQL Row-Level Security (RLS) policies. Precise GPS coordinates are harvested exclusively upon active crash triggers or explicit user consent, eliminating the risk of unmonitored background surveillance.

VII. CONCLUSION AND FUTURE WORK

In this work, we developed and field-benchmarked RESQONE-AI+, an integrated edge-cloud emergency coordination ecosystem engineered to conquer the physiological Golden Hour. By synthesizing edge-computed crash telemetry (≥ 4.0 G), transparent multilingual NLP triage, real-time hospital AVS inventory mapping, and localized blood-courier logistics, the platform reduced end-to-end emergency dispatch latency by 88.58% (from 18.4 down to 2.1 minutes). These findings demonstrate that edge-computed intelligence and explainable decision pipelines can bridge the logistical divide in resource-constrained emergency medical systems.

Future work will advance this framework along three research trajectories:

- Device-to-device ad-hoc mesh networking using BLE and Wi-Fi Direct to relay emergency distress packets during total telecommunications collapses.
- Privacy-preserving federated edge learning to continuously retrain diagnostic triage weights across local hospital nodes without centralizing confidential patient transcripts.
- Cryptographically auditable smart-contract ledgers for verifying volunteer first-responder credentials and emergency response times.

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